

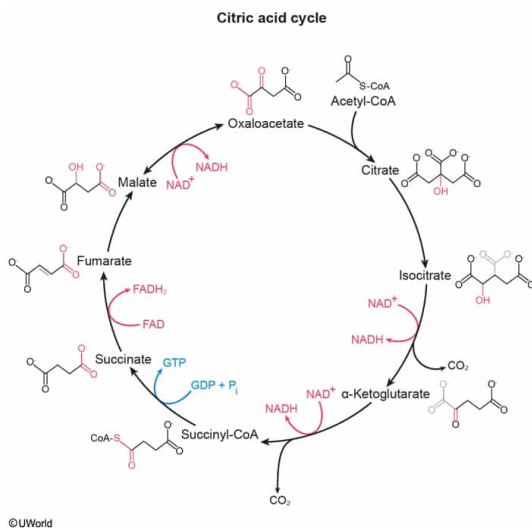
Which of the following steps in the citric acid cycle does NOT produce reduced electron carriers?

- ☐ A. Conversion of isocitrate to  $\alpha$ -ketoglutarate [10%]
- ☐ B. Conversion of  $\alpha$ -ketoglutarate to succinyl-CoA [8%]
- ✓ ☒ C. Conversion of succinyl-CoA to succinate [64%]
- ☐ D. Conversion of succinate to fumarate [16%]

Correct

65% answered correctly

### Explanation:



The **citric acid cycle** is an **oxidative process**. The **oxidation state** of carbon can be estimated by the number of **oxygen or sulfur bonds** and the number of **C=C double bonds**. Increasing the number of these bonds increases the overall oxidation state. When one molecule is oxidized, another molecule must be reduced, and vice versa. Therefore, each of the four oxidative steps in the cycle yield reduced electron carriers, either in the form of NADH or  $\text{FADH}_2$ .

The oxidative steps are:

1. Conversion of isocitrate to  $\alpha$ -ketoglutarate (**Choice A**). In this reaction, a single bond to oxygen (hydroxyl group) is oxidized to a ketone with two bonds to oxygen, and a carboxyl group leaves as  $\text{CO}_2$ . One NADH molecule is produced in this step.
2. Conversion of  $\alpha$ -ketoglutarate to succinyl-CoA (**Choice B**). In this step, the ketone that was formed in the first step is further oxidized by formation of a bond to sulfur, and another carboxyl group leaves as  $\text{CO}_2$ . Another NADH molecule is formed during this step.
3. Conversion of succinate to fumarate (**Choice D**). In this step, a carbon is oxidized by forming a double bond with another carbon. Oxidation steps of this nature typically use FAD as a cofactor, and this step produces one  $\text{FADH}_2$  molecule as a reduced electron carrier.
4. Conversion of malate to oxaloacetate. In this step, a hydroxyl group is oxidized to a ketone, and a molecule of NADH is formed in the process.

Conversion of **succinyl-CoA to succinate** does **not involve oxidation** because the C–S bond

that breaks and the C–O bond that forms both have the same oxidation state. With no oxidation reaction, no reduction of electron carriers can occur. However, this conversion is still a high-energy step as it produces guanosine triphosphate (GTP), which is equivalent in energy to ATP.

**Educational objective:**

The citric acid cycle includes four oxidative steps. Each oxidative step is coupled to the reduction of an electron carrier, either  $\text{NAD}^+$  to NADH or FAD to  $\text{FADH}_2$ .

**Time spent:**0 sec

**Subject:**  
Biochemistry

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**System:**  
Metabolic Reactions